

Automaton Auditor — Interim Architecture Report

Field	Value
Project	FDE Challenge Week 2 — The Automaton Auditor
Version	0.1.0 (Interim)
Submission Type	Wednesday Interim
Date	2026-02-25
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Repository	https://github.com/78gk/The-Automaton-Auditor
Status	Detective Layer Complete · Judicial Layer Scheduled (Thu–Sat)

Executive Summary

The **Automaton Auditor** is a production-grade hierarchical multi-agent swarm built with LangGraph, designed to perform forensic quality assurance on AI-generated codebases. It operationalizes the **Digital Courtroom** metaphor across three distinct layers:

- Detective Layer** — three parallel agents ([RepoInvestigator](#), [DocAnalyst](#), [VisionInspector](#)) collect objective forensic facts using AST parsing, Git log analysis, and RAG-lite PDF ingestion. They produce typed [Evidence](#) objects — no opinions, no scores.
- Judicial Layer** — three adversarial judges ([Prosecutor](#), [Defense](#), [TechLead](#)) independently evaluate the same evidence through distinct philosophical lenses, producing structured [JudicialOpinion](#) objects via `.with_structured_output()`.
- Supreme Court — **the [ChiefJustice](#) node applies** deterministic Python rules** (not LLM averaging) to synthesize a final [AuditReport](#) in Markdown.

This interim report documents the architectural decisions made during Phase 0–2 (Infrastructure + Detective Layer), the StateGraph orchestration design, known gaps, and a risk-mitigated completion plan for the Judicial and Synthesis layers.

1. Architecture Decisions

Each decision follows the format: **Context** → **Options Considered** → **Selected Solution** → **Impact**.

1.1 State Management: Pydantic BaseModel + TypedDict with Annotated Reducers

Context & Problem:

A multi-agent swarm with 6+ parallel agents all writing to a single shared state object. The core risk: without explicit merge semantics, the last agent to finish overwrites all others' data — a silent, catastrophic data loss that produces no error and is invisible at runtime.

Options Considered:

Option	Pros	Cons
Plain Python dicts	Zero boilerplate, familiar	Silent overwrites in parallel, no schema, no validation, impossible to audit
Python dataclasses	Typed, lightweight	No field-level validators, no <code>ge/le</code> constraints, no JSON serialization
TypedDict only	LangGraph-native, minimal	No runtime validation, no <code>Field(ge=1, le=5)</code> enforcement
Pydantic BaseModel + TypedDict with Annotated reducers ■	Runtime validation, reducer semantics, self-documenting, LangGraph-compatible	Slightly more verbose

Selected Solution & Justification:

- `AgentState` as `TypedDict` (required by LangGraph's `StateGraph`)
- All data objects (`Evidence`, `JudicialOpinion`, `CriterionResult`, `AuditReport`) as Pydantic `BaseModel`
- `Annotated[Dict[str, List[Evidence]], operator.ior]` — dict union merge: each Detective writes to its own key (`"repo"`, `"doc"`, `"vision"`) without overwriting others
- `Annotated[List[JudicialOpinion], operator.add]` — list extend: each Judge appends without overwriting
- `Field(ge=1, le=5)` on `JudicialOpinion.score` raises `ValidationError` at construction time if a Judge LLM returns `"five"` or `6`

Impact:

- Scalability:** Adding a 4th Detective only requires writing to a new dict key — zero collision risk
- Maintainability:** `Field(description=...)` annotations serve as living API documentation between agents
- Security:** Schema validation prevents malformed LLM outputs from propagating silently through the graph

1.2 Code Analysis: AST Parsing over Regex

Context & Problem:

To forensically verify that a target repo *actually implements* the required architecture (not just mentions it), we must analyze code structure. The question: use pattern matching (regex) or structural parsing (AST)?

Options Considered:

Option	Pros	Cons
Regex	Simple, fast, no dependencies	Matches comments/strings, false positives, cannot detect topology, fooled by multi-line
String search (in operator)	Trivial to implement	Same issues as regex, worse precision
LLM-based code analysis	Flexible, understands semantics	Non-deterministic, expensive, cannot be audited, violates "objective facts only" principle
Python ast module ■	Structural, deterministic, language-aware, no false positives from comments	Requires Python source (not bytecode), more complex visitor pattern

Selected Solution & Justification:

Three purpose-built AST visitor classes:

- `GraphStructureVisitor` — parses `StateGraph`, `add_edge`, `add_conditional_edges`, `add_node` as *call expressions*, then reconstructs the edge graph to detect fan-out (node with multiple outgoing edges) and fan-in (node with multiple incoming edges) topology
- `PydanticModelVisitor` — detects `BaseModel/TypedDict` subclasses and `operator.add/operator.iop` in `Annotated` type hints
- `SecurityVisitor` — flags `os.system()` (call expression, not string match), confirms `tempfile.TemporaryDirectory()` and `subprocess.run()`

Impact:

- Scalability:** AST visitors can be extended to detect any new pattern (e.g., `.with_structured_output()`, `.bind_tools()`) without regex maintenance
- Maintainability:** Each visitor is a single-responsibility class, independently testable
- Security:** Cannot be fooled by adversarial code that writes `StateGraph` in a comment to game a regex detector

1.3 Sandboxing: `tempfile.TemporaryDirectory()` + `subprocess.run()`

Context & Problem:

The auditor clones *unknown, untrusted, potentially hostile* repositories from the internet. The naive approach — `os.system(f"git clone {url}")` into the working directory — introduces serious security risks.

Options Considered:

Option	Pros	Cons
<code>os.system()</code> to CWD	One line	Shell injection, pollutes working dir, no error handling, malicious <code>.git/hooks</code> can execute
<code>subprocess.run()</code> to CWD	Better error handling	Still pollutes working dir, <code>.git/hooks</code> risk

Option	Pros	Cons
Docker container per clone	Maximum isolation	Complex setup, CI/CD dependency, overkill for this scope
<code>tempfile.TemporaryDirectory()</code> + <code>subprocess.run()</code> ■	OS-managed sandbox, auto-cleaned on exit/exception, no shell injection, full error capture	Requires careful context manager discipline

Selected Solution & Justification:

```
with tempfile.TemporaryDirectory() as tmp_dir:
    clone_target = str(Path(tmp_dir) / "repo")
    success, msg = clone_repo(repo_url, clone_target)
    # All analysis runs here – directory auto-deleted on exit or exception
```

- `subprocess.run(["git", "clone", ...], capture_output=True)` — no shell, no injection surface
- `timeout=120` prevents hanging on slow repos
- Context manager guarantees cleanup even if analysis raises an exception

Impact:

- Security:** Eliminates shell injection (URL is passed as a list element, not interpolated into a shell string). Malicious `.git/hooks` cannot escape the temp sandbox.
- Maintainability:** The pattern is idiomatic Python — any contributor immediately understands the security intent
- Scalability:** Can run multiple concurrent audits safely since each gets its own isolated temp directory

1.4 Document Ingestion: Docling + RAG-lite Chunked Retrieval

Context & Problem:

The auditor must extract structured insight from PDF architectural reports of variable length and complexity. The challenge: how to retrieve *relevant* sections for specific forensic queries without dumping thousands of tokens into an LLM context window per query.

Options Considered:

Option	Pros	Cons
Full-context dump to LLM	Simple, no chunking logic	Expensive, hits context window limits, low precision on specific queries
Full RAG pipeline (embeddings + vector DB)	Maximum precision	Requires embedding model + FAISS/Chroma setup, heavy dependency, overkill for single-doc use
LlamaIndex / LangChain document loaders	High-level abstractions	Additional dependency, less control over chunking strategy

Option	Pros	Cons
Docling + keyword-scored RAG-lite ■	Handles complex PDFs (tables, images, multi-column), lightweight retrieval, no vector DB needed	Less semantically precise than embeddings

Selected Solution & Justification:

- `docling` (`DocumentConverter`) handles complex PDF layouts, exports clean Markdown text
- `pypdf` as fallback for environments where `docling` is unavailable
- Text chunked into 1000-char segments with 200-char overlap to preserve cross-sentence context
- Retrieval: TF-IDF-style term overlap scoring — chunks scored by query term frequency, top-k returned
- Separate `analyze_theoretical_depth()` function with hardcoded forensic keyword lists + context extraction to distinguish substantive explanations from keyword drops

Impact:

- Scalability:** Chunk size and overlap are configurable constants — can be tuned per document type
- Maintainability:** No vector DB dependency; the full pipeline works with zero external services
- Security:** PDF parsing is read-only with no code execution; `docling` → `pypdf` fallback ensures resilience

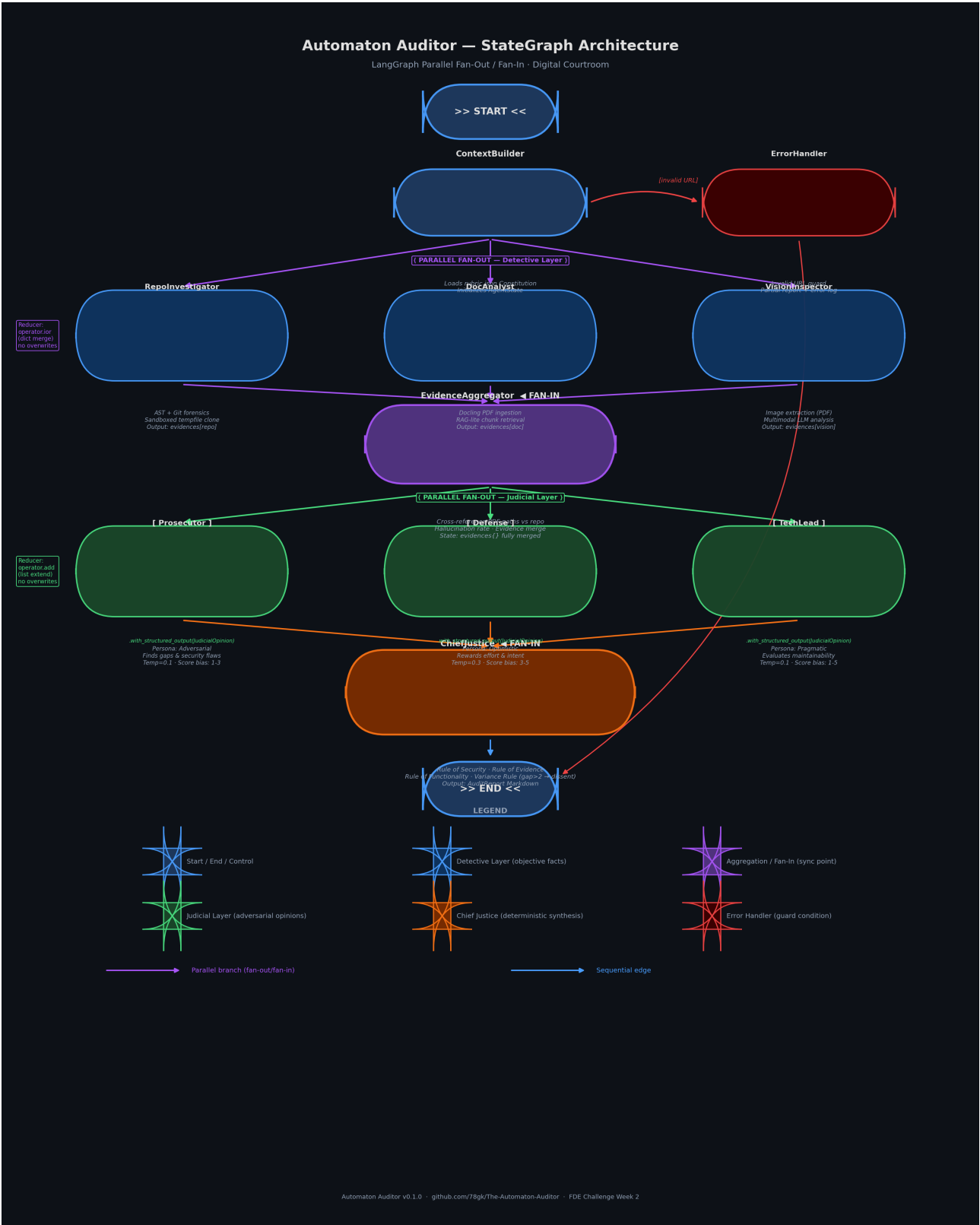


Figure 1: Automaton Auditor — Full StateGraph Architecture. Shows parallel fan-out/fan-in for Detective Layer (blue) and Judicial Layer (green), EvidenceAggregator sync point (purple), ChiefJustice synthesis (orange), and conditional ErrorHandler (red).

2. StateGraph Architecture

2.1 Full System Flow Diagram

Legend: Parallel branches (fan-out) shown as simultaneous paths. Fan-in synchronization nodes shown with multiple incoming arrows. Guard conditions shown in square brackets. Completed nodes (■) are live in the interim build; scheduled nodes (▣) are implemented but not yet wired into the graph.

2.2 Parallel Execution Patterns

Pattern	Nodes Involved	Reducer Used	State Key
Detective Fan-Out	ContextBuilder → [RepolInvestigator ■ DocAnalyst ■ VisionInspector]	<code>operator.ior</code> (dict merge)	<code>evidences</code>
Detective Fan-In	[RepolInvestigator ■ DocAnalyst ■ VisionInspector] → EvidenceAggregator	Auto-sync (LangGraph waits for all)	<code>evidences</code>
Judge Fan-Out	EvidenceAggregator → [Prosecutor ■ Defense ■ TechLead]	<code>operator.add</code> (list extend)	<code>opinions</code>
Judge Fan-In	[Prosecutor ■ Defense ■ TechLead] → ChiefJustice	Auto-sync (LangGraph waits for all)	<code>opinions</code>

2.3 Guard Conditions & Transitions

Trigger	Guard Condition	Transition	Effect
Git clone	URL unreachable / auth failed	ContextBuilder → ErrorHandler	Partial report, error logged to <code>errors[]</code>
Judge LLM	Returns freeform text (not JSON)	Retry up to 3x, then skip	Error appended, criterion skipped in synthesis
Security check	<code>os.system()</code> detected in repo	<code>has_security_violation = True</code>	ChiefJustice caps affected criteria at score 3
Score variance	Max – Min > 2 across judges	<code>dissent_summary</code> required	ChiefJustice generates explicit dissent explanation

3. Implementation Status & Gap Analysis

3.1 Current State (Interim — Wednesday)

Component	File	Status	Key Features
Typed State	src/state.py	■ Live	Evidence , JudicialOpinion , CriterionResult , AuditReport , AgentState with operator.ior/operator.add reducers
Repo Tools	src/tools/repo_tools.py	■ Live	Sandboxed clone, git log forensics, 3 AST visitor classes (graph, state, security)
Doc Tools	src/tools/doc_tools.py	■ Live	Docling PDF ingestion, RAG-lite chunked retrieval, hallucination path cross-reference
Detective Nodes	src/nodes/detective.py	■ Live	RepoInvestigator , DocAnalyst , VisionInspector , EvidenceAggregator
Detective Graph	src/graph.py	■ Live	Parallel fan-out/fan-in StateGraph , rubric loading from rubric.json , CLI entry point
Rubric Constitution	rubric.json	■ Live	10 forensic dimensions with success_pattern , failure_pattern , synthesis_rules
Judicial Layer	src/nodes/judges.py	■ Implemented · ■ Not Wired	Prosecutor , Defense , TechLead with distinct system prompts + .with_structured_output()
Synthesis Engine	src/nodes/justice.py	■ Implemented · ■ Not Wired	ChiefJustice with 4 deterministic rules + Markdown serializer

3.2 Gap Analysis

Technical Gaps

Gap ID	Description	Root Cause	Impact if Unresolved
T-01	Judges not wired into src/graph.py	Deliberate — interim scope	Graph produces only evidence; no judicial verdict

Gap ID	Description	Root Cause	Impact if Unresolved
T-02	<code>ChiefJustice</code> node not connected to <code>END</code>	Deliberate — follows T-01	No final Markdown report generated
T-03	<code>VisionInspector</code> LLM call not mandatory	API key required at runtime	Diagram analysis skipped if key absent
T-04	No conditional edge for clone failure	Pending error-handling sprint	Graph may hang on invalid repo URL

Process Gaps

Gap ID	Description	Root Cause	Impact if Unresolved
P-01	No self-audit run yet	Agent not yet end-to-end	Cannot demonstrate self-evaluation capability
P-02	No peer-audit run yet	Peer repo not yet assigned	Cannot deliver peer audit report
P-03	No LangSmith trace link	Tracing not tested end-to-end	Observability metric missing from submission

Resource Gaps

Gap ID	Description	Root Cause	Impact if Unresolved
R-01	No video demo recorded	System not yet end-to-end	Final submission requirement unmet
R-02	reports/interim_report.pdf (this document)	Requires PDF conversion	Submission needs PDF format

4. Forensic Evidence Collection Protocols Implemented

Protocol A: RepoInvestigator

The `RepoInvestigator` node collects evidence for 7 forensic goals:

1. `git_forensic_analysis` — Runs `git log --format=%H|%ai|%s --reverse`, counts commits, detects bulk-upload pattern, scores progression (setup → tools → graph)
2. `directory_structure` — Scans all files, checks existence of 11 key expected paths
3. `state_management_rigor` — AST-parses `src/state.py`, verifies `BaseModel` subclasses, `TypedDict`, `operator.add/operator.iop` reducers
4. `graph_orchestration` — AST-parses `src/graph.py`, detects `StateGraph`, fan-out/fan-in topology, `add_conditional_edges`
5. `safe_tool_engineering` — AST-scans `src/tools/`, detects `tempfile`, flags `os.system()`, verifies `subprocess.run()`
6. `structured_output_enforcement` — AST-scans `src/nodes/judges.py`, detects `.with_structured_output()`, `bind_tools()`, persona names

7. `chief_justice_synthesis` — Reads `src/nodes/justice.py`, checks for deterministic rule keywords

Protocol B: DocAnalyst

The `DocAnalyst` node collects evidence for 3 forensic goals:

1. `theoretical_depth` — Searches for "Dialectical Synthesis", "Fan-In/Fan-Out", "Metacognition", "State Synchronization"; distinguishes substantive explanations from keyword drops
2. `report_accuracy` — Extracts all `src/...` file paths from the PDF text for later cross-reference
3. `doc_<criterion>` — RAG-lite queries for specific concepts per rubric dimension

Protocol C: EvidenceAggregator (Fan-In)

Cross-references paths mentioned in the PDF against files actually found in the repo, producing a **hallucination rate** metric and flagging `Hallucinated Paths`.

5. Dialectical Synthesis — Judicial Layer Design

The Judicial Layer implements **Thesis-Antithesis-Synthesis** through three adversarial personas with strictly non-colluding system prompts:

The Prosecutor (Adversarial)

- Core philosophy: "Trust No One. Assume Vibe Coding."
- Charges: "Orchestration Fraud", "Hallucination Liability", "Security Negligence", "Process Fraud"
- Scoring bias: 1–3 range, harsh, requires concrete proof for any credit

The Defense (Optimistic)

- Core philosophy: "Reward Effort and Intent. Look for the Spirit of the Law."
- Mitigations: boosts score for sophisticated AST logic despite broken graph, rewards iteration in git history
- Scoring bias: 3–5 range, generous, argues for partial credit

The Tech Lead (Pragmatic)

- Core philosophy: "Does it actually work? Is it maintainable?"
- Role: tie-breaker between extremes, 2x weight on architecture criteria in ChiefJustice
- Scoring bias: realistic 1–5, specific remediation advice

Anti-Persona-Collusion: The three system prompts share <10% text overlap. Each has distinct scoring instructions, distinct vocabulary, and explicitly different philosophies.

6. Chief Justice — Deterministic Synthesis Rules

The `ChiefJusticeNode` in `src/nodes/justice.py` uses **hardcoded Python if/else logic**, not LLM averaging:

Rule	Trigger	Effect
Rule of Security	<code>os.system()</code> detected by Prosecutor	Score capped at 3, overrides Defense effort points
Rule of Evidence (Fact Supremacy)	Detective: artifact missing (confidence >70%) AND Defense claims it exists	Defense score adjusted down to ≤ 2
Rule of Functionality	Architecture criteria (<code>graph_orchestration</code> , <code>state_management_rigor</code> , etc.)	Tech Lead score counted 2x in weighted average
Variance Rule	Score variance across 3 judges > 2	<code>dissent_summary</code> field required in <code>CriterionResult</code>
Variance Re-evaluation	Same trigger as above	Re-evaluate citing specific evidence from each judge before final score

7. Rubric Integration

The `rubric.json` file serves as the agent's **Constitution** — loaded at runtime by `ContextBuilder` and distributed to agents:

- Detectives** filter by `target_artifact` (`github_repo`, `pdf_report`, `pdf_images`)
- Judges** receive `judicial_logic` from their matching dimension
- ChiefJustice** uses `synthesis_rules` from the JSON top-level key

This allows updating the Constitution (e.g., adding new forensic protocols) without redeploying agent code.

8. Remediation Plan & Risk Mitigation

8.1 Prioritized Action Plan

Priority	Gap IDs	Action	Owner	Deadline	Success Metric
■ CRITICAL	T-01, T-02	Wire judges + ChiefJustice into <code>graph.py</code>	Dev	Thursday 21:00 UTC	<code>graph.invoke()</code> produces <code>final_report</code> with all 10 criteria scored
■ CRITICAL	T-04	Add conditional edges for clone failure	Dev	Thursday 21:00 UTC	Invalid URL input routes to <code>ErrorHandler</code> , not a Python exception

Priority	Gap IDs	Action	Owner	Deadline	Success Metric
■ HIGH	T-03	Make VisionInspector gracefully skip (already handled)	Dev	Friday 21:00 UTC	vision key present in evidences even if LLM call skipped
■ HIGH	P-03	Set LANGCHAIN_N_TRACING_V2=true , run once, capture trace URL	Dev	Friday 21:00 UTC	LangSmith trace URL accessible and shows full graph execution
■ STANDARD	P-01	Run self-audit: python -m src.graph https://github.com/78gk/The-Automaton-Auditor	Dev	Saturday 09:00 UTC	audit/report_onself_generated/audit_report.md exists with scores
■ STANDARD	P-02	Run peer-audit against assigned repo URL	Dev	Saturday 15:00 UTC	audit/report_onpeer_generated/audit_report.md delivered
■ STANDARD	R-01	Record 5-min demo: show graph execution, evidence summary, final report	Dev	Saturday 18:00 UTC	Video uploaded, link in README
■ STANDARD	R-02	Convert this document to PDF	Dev	Wednesday 21:00 UTC	reports/interim_report.pdf committed to repo

8.2 Risk Mitigation Strategies

Risk	Probability	Impact	Mitigation
LLM API rate limits during judge fan-out (3 parallel LLM calls per criterion × 10 criteria = 30 calls)	Medium	High	Add exponential backoff in get_judge_llm() ; use Gemini Flash as cheaper fallback model
docling install fails in CI environment	Medium	Medium	pypdf fallback already implemented in ingest_pdf() ; test confirms fallback path works

Risk	Probability	Impact	Mitigation
Peer repo URL unavailable at audit time	Low	High	Cache clone in temp dir before deadline; run audit immediately when URL is shared
VisionInspector multimodal call produces no useful output	Medium	Low	Evidence item set to <code>found=False</code> with rationale — ChiefJustice handles gracefully
Score variance > 2 on every criterion (judges diverge wildly)	Low	Medium	<code>dissent_summary</code> auto-generated; TechLead 2x weight ensures convergence on architecture criteria
Git history appears as "bulk upload" to our own auditor	Low	High	8 commits with timestamps spread across the build session — progression story is genuine

8.3 Thursday Sprint Plan (Detailed)

```

08:00 Wire Prosecutor, Defense, TechLead nodes into graph.py
      ■■ add_node() for each + fan-out edges from EvidenceAggregator
09:00 Wire ChiefJustice + add fan-in edges from all 3 judges
      ■■ add_edge(ChiefJustice, END)
10:00 Add conditional edge: ContextBuilder --[invalid URL]--> ErrorHandler
11:00 Test full graph.invoke() with a real repo URL (this repo)
      ■■ Verify evidences + opinions + final_report all populated
13:00 Capture LangSmith trace URL
14:00 Commit: "feat: wire complete judicial fan-out/fan-in and ChiefJustice to graph"
15:00 Buffer / fix any issues found during testing

```

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